

Temperature-Aware Farm System

Fundamentals of AI Using Agriculture Data Set

Student Details

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1. Introduction

Agriculture plays a vital role in food security and economic development. Farmers often suffer losses due to crop spoilage caused by high temperatures and inadequate storage facilities. Artificial Intelligence can help predict spoilage risks and support timely decision-making.

This project develops a Temperature-Aware Farm System that predicts spoilage risk using machine learning techniques.

2. Problem Statement

High temperatures and poor storage conditions can lead to significant crop spoilage. Farmers need an intelligent system that can identify spoilage risks before losses occur.

3. Objectives

- Analyze agricultural data related to temperature and storage.
- Develop a machine learning model to predict spoilage.
- Identify important factors affecting spoilage.
- Assist farmers in making informed decisions.

4. Dataset Description

The dataset contains the following attributes:

- Record_ID
- Date
- Location
- Crop_Type
- Temperature_C

- Storage_Available
- Spoilage_Observed

Additional features were generated:

- Relative_Humidity_%
- Heatwave_Alert
- Days_In_Storage
- Transport_Delay_Hours

5. Methodology

Data Preprocessing

- Data loading using Pandas
- Feature engineering
- Label encoding of categorical variables

Machine Learning Model

- Supervised Learning
- Random Forest Classifier

Evaluation

- Accuracy Score
- Confusion Matrix
- Feature Importance Analysis

6. Results

The Random Forest model successfully predicts spoilage risk using temperature, storage conditions, and other environmental factors.

Feature importance analysis showed that temperature and storage-related factors significantly influence spoilage prediction.

7. Benefits

- Early spoilage prediction
- Reduced agricultural losses

- Better storage management
- Improved decision-making for farmers

8. Future Scope

Future improvements may include:

- IoT sensor integration
- Real-time weather forecasting
- Mobile application support
- Automated farmer alerts

9. Conclusion

The Temperature-Aware Farm System demonstrates how Artificial Intelligence can be applied in agriculture to reduce spoilage losses and improve farm productivity. The project highlights the effectiveness of machine learning in supporting agricultural decision-making.

References

1. Python Documentation
2. Scikit-Learn Documentation
3. Pandas Documentation
4. IIT Ropar ANNAM.AI Course Material